Table 6 Species group breakdown of recent industrial trawl catches as reported in Yearbook of Fisheries Statistics of Bangladesh (2020).

Type of Fishing	No. of active Trawler	No. of Unit (Trawl Net)	Catch in 2019-20 (MT)			
			Shrimp	Hilsa	Other Fish	Total
Shrimp Trawler	30	90	1,457	0	2,749	4,206
Fish Trawler	190	570	979	9,616	100,553	111,148
Total	220	660	2,436	9,616	103,302	115,354

Species group breakdown of other fish category (MT)								
Sardine	Bombay Duck	Indian Salmon	Pomfret	Jewfish	Catfish	Sharks Skates Rays	Other Marine Fish	Total
16,154	6,494	0	1,205	6,271	5,223	602	67,353	103,302

Current management measures and arrangements

Licensing

A valid fishing license, issued by the Marine Fisheries Office (MFO), is required for industrial marine fishing vessels. Licenses are renewed biannually and require possession of vessel registration and valid inspection certificates from Mercantile Marine Office (MMO).

Government has the authority to regulate number of fishing vessel license, as provided for in Marine Fisheries Act 2020 section 4 below. The intention is to reduce over-capacity while encouraging modernisation of the industrial trawler fleet.

"Section 4 – Determination of Classes and Numbers of Fishing Vessels: - In order to maintain conservation, management, sustainable and extractable stock of fisheries resources and in need of development planning Government, by notification in the Government Gazette, for providing license in the marine area of fish extraction or in the deep sea, from time to time, may determine the number and classification of the vessels."

Enforcement agencies

Enforcement of fisheries management measures by the GoB includes the DoF, Bangladesh Coast Guard (BCG) and Bangladesh Navy (BN). However implementation and compliance is very weak. There is no effective co-management mechanism or process resulting in a very top-down approach and limited legitimacy in the eyes of fishers. The existing industrial and artisanal fisheries associations are a potential institutional base for a future co-management system.

Rules and regulations

The management measures in effect for industrial fisheries under the *Marine Fisheries Act 2020* and *Rules* are:

- Licensing
  - All industrial trawlers are required to have an industrial fishing vessel license which is to be renewed biannually

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- All industrial trawlers and mechanized fishing boats are required to have vessel registration and sea worthiness certificate (Certificate of Inspection) from MMO before applying for a fishing license or biannual renewal.

- Area of operations
  - All industrial vessels must carry and operate Automatic Identification System (AIS) throughout entire fishing trips
  - From the start of the 2021-22 fishing season or when the legal provisions and infrastructure is ready, whichever is earlier, vessels must carry and operate Vessel Monitoring System (VMS) throughout entire fishing trips
  - Vessels fishing under an industrial fishing vessel license are required to fish beyond (seaward of) the inshore limit line defined by the coordinates given below

Point	E Long (DMS)	N Lat (DMS)	E Long (DD)	N Lat (DD)
1	89°13'30"	21°11'00"	89.225°	21.183°
2	89°32'00"	21°25'00"	89.533°	21.417°
3	89°40'00"	21°25'00"	89.667°	21.417°
4	89°40'00"	21°18'00"	89.667°	21.300°
5	90°30'00"	21°06'00"	90.500°	21.100°
6	91°16'30"	21°00'00"	91.275°	21.000°
7	91°32'00"	21°06'00"	91.533°	21.100°
8	91°45'00"	21°06'00"	91.750°	21.100°
9	92°09'00"	20°25'30"	92.150°	20.425°

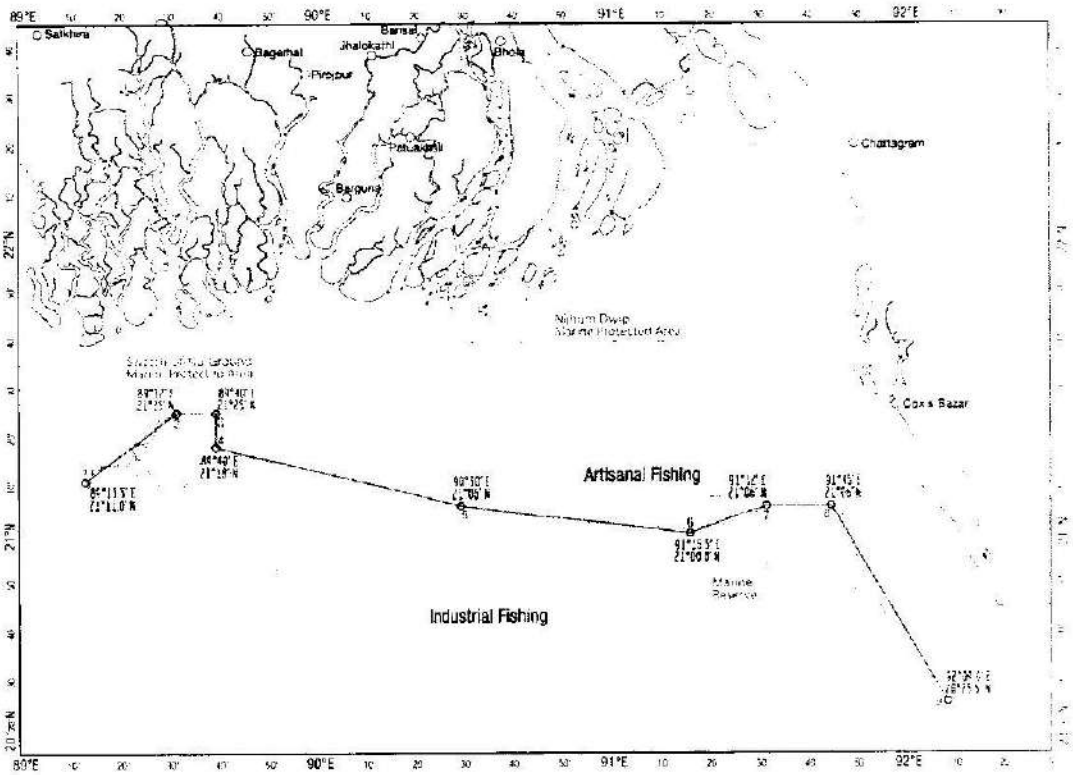

Figure 5 Spatial restrictions for industrial trawling

- GoB has declared a continental shelf Marine Protected Area (Marine Reserve) between Middle Ground and South Patches encompassing 698 sq. km. (204 sq. nm.) to protect spawning nursery habitat for marine shrimp and fish.

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- GoB has declared 3188 sq. km. of Marine Protected Area south of Nijhum Dwip to conserve marine and coastal fisheries resources.
- GoB has restricted collection of fry of fish, or PL of shrimp and prawns of any kind in the estuary and coastal waters of Bangladesh since 2000.
- GoB encourages industrial fishing fleet to conduct exploratory and/or developmental fishing outside 500 m isobaths within EEZ.

- Closed season
  - Fishing by all types of fishing vessels is banned from 20 May to 23 July each year to promote the conservation of marine fisheries resources within the EEZ of Bangladesh.
- Trip duration
  - Freezer trawlers are permitted a maximum trip duration of 30 days
  - Non-freezer trawlers are permitted a maximum trip duration of 14 days
- Sailing permission and Arrival report
  - All licensed fishing vessels shall have sailing permission
  - At the end of fishing all fishing vessels shall report 24 hours prior to arrive at port
  - After receiving arrival report, entitled officer shall inspect amount and type of harvested fish
- By-catch
  - Shrimp trawlers must land, not discard, finfish
  - Shrimp trawlers must have at least 30% fin-fish in the total catch
  - Control of mesh size:
    - cod-end of shrimp trawl nets must be 45 mm or larger;
    - cod-end of fin-fish trawler net must be 60 mm or larger;
  - Shrimp trawlers must use Turtle Excluder Device (TEDs)
  - All trawlers shall use specified Bycatch Reduction Devices (BRDs), including TEDs when available.

Hilsa Fisheries Management Plans

Hilsa is both economically and culturally important in Bangladesh and supports a vital and closely managed freshwater fishery.

The HFMP has been in effect since 2003 resulting in successful conservation of Hilsa while allowing a 100% increase in production over the last 16 years and supporting the livelihoods of around 0.5 million fishers. The main element of the strategy is spatial protection of four critical spawning grounds and five Hilsa and juvenile Hilsa or 'jatka' sanctuary areas/nursery grounds through seasonal fishing bans (Table 7). A compensation package to affected fishers under a GoB safety net arrangement encourages high compliance.

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Table 7 Fishing ban areas and periods specified under the Hilsa Fisheries Management Plan

Ban Area	Locality	Ban period
North East	Mayani Point, Mirsarai, Chittagong in the northeast (91°32.15' E and 22°42.59' N)	4 days before and 17 days after the full moon including the day of full moon, that is, total 22 days of the first full moon in the Bengali month of Ashwin each year
North West	Paschim Syed Awlia Point, Tajmuddin, Bhola in the northwest (90°40.58' E and 22°31.16' N)	
South East	North Kutubdia Point, Kutubdia, Cox's Bazar in the southeast (90°52.51' E and 21°55.19' N)	
South West	Lata Chapali Point, Kalapara, Patuakhali in the southeast (90°12.59' E and 21°47.56' N)	

The successful implementation of the HFMP demonstrates the feasibility and effectiveness of this fisheries management plan, underpinned by scientific information, in establishing appropriate catch and/or effort limitations, preventing overfishing, and rebuilding the Hilsa stock. It is clear that an unregulated expansion of Hilsa fishing in marine waters is underway leading to the growing marine catch of Hilsa, particularly by midwater trawlers.

To ensure the continued successful management of this stock, based on the existing HFMP, it is critical that targeted industrial trawling for Hilsa is to be eliminated and measures must be included to ensure Hilsa bycatch is kept to the minimum possible level in the industrial sector (Table 8).

Table 8 Management measures to limit Hilsa catch in Industrial fisheries

Measure	Notes
BRD	Midwater trawlers shall use trawl BRD when available, sized to exclude Hilsa, in addition to TED
"Move-on" rule	When Hilsa catch exceeds 5% of the catch in a tow by weight, the vessel must relocate to avoid the area
Fishing closures	Time and area closures may be introduced when specific areas are affected by high bycatches of Hilsa.
Bycatch limit	Hilsa landings are limited to a maximum bycatch of 10%* by weight of total catch, for all industrial trawlers.

*Bycatch limit will be executed after development of specific BRDs for Hilsa

Monitoring, control and surveillance

Overview of fisheries MCS

The ultimate goal of fisheries management is to maximize the sustainable benefits and economic return from the country's territorial waters and exclusive economic zone. The success of a FMP depends on it being based on adequate information and sound decision-making, and being implemented through a strong and cost-effective Monitoring, Control and Surveillance (MCS) system. Such a system is an integrated information collation, rule-making and enforcement system providing tools for implementation of policies, strategies and frameworks for fisheries management and other aspects of ocean and environmental governance. MCS is critical for the implementation of a successful fisheries management strategy and plan that renders a fisheries management regime effective and sustainable.

Implementation of MCS

Bangladesh currently has the following MCS processes:

- Industrial vessels must land catch at designated ports in the presence of MFO officials authorized to perform inspections and verifying paper based daily fish log with landed catch;

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- Industrial vessels must use only fishing gears meeting the regulated specifications
- Industrial vessels require sailing permission (SP) from MFO before departing for fishing after submitting
    - proof of valid license
    - stipulated fee
    - catch log sheet of previous trip
- Vessels are inspected regularly or randomly by authorized personnel of MFO of DoF before and after trip at port or as shore based inspection at Marine Fisheries Surveillance Check posts (MFSCs).
- At-sea monitoring of industrial fishing vessels is carried out by patrolling vessels of BN and BCG.
    - Patrolling vessels may carry DoF Officers as observers or in joint operations
    - BCG conducts patrols including MCS actions within the 40 m depth contour
    - BN conducts patrols including MCS actions in offshore EEZ waters (seaward of 40 m depth contour)
- IUU catch certification for traceability issued by MFO for compliance with EU requirements.

New management measures and MCS activities

Many of the above measures and regulations are not implemented effectively or comprehensively, primarily due to limited human and logistical resources within MFO. During the span of this plan it is intended to implement a number of new capabilities and activities.

- Establish a Joint Monitoring Centre (JMC) in the Chattogram port area interagency coordination in MCS under Standard Operating Procedures (SOP) for implementation of the Marine Fisheries Act and rules in the EEZ of Bangladesh.
- Establish operational electronic fishing vessel licensing system including IT infrastructure, training, capacity building & reporting. Data of all registered fishing vessels incorporated in the electronic system.
- Establish infrastructure for data collection by at-sea observers with required mobile application, virtual server, database, and back-end data management software, with user manuals and training, to support monitoring fisheries activities and reporting to competent authority.
- Establish new MFSC at strategic locations in coastal districts, including facilities for high speed patrol vessels to conduct joint monitoring and surveillance program with maritime cooperating agencies.
- Establish a new VMS through a VMS provider and require VMS installation as a condition of license for all industrial fishing vessels. VMS will support recording and displaying vessel position and track data, pattern of fishing activities, use of legitimate gear, catch transhipment at sea, and e-log fish catch report. VMS will integrate with AIS and establish connection with JMC for strengthening MCS activities through communication & coordination with participating marine domains for inspection and enforcement. But the legal regime requires provision by vessel owners for installing VMS transponder and borne airtime fees for satellite for sustainability question.
- Establish an electronic catch documentation scheme through e-reporting of catch and e-logbooks to increase transparency of fish moving through the supply chain.

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Major threats and high priority challenges

Conflicts between artisanal and industrial operators

The industrial and artisanal fishers share the same fisheries resources in many cases. Overcapacity and intense fishing pressure in both artisanal and industrial sectors has led to increasing competition and conflicts for the depleted resources. Although regulations to separate the sectors at the 40 m contour are in place for many years, there is extensive infringement of inshore fishing zones by industrial trawlers and to a lesser extent by artisanal into offshore fishing areas (Figure 6). This is expected to worsen as the uncontrolled and unabated increase in numbers of artisanal boats increases the pressure on fish resources in marine waters.

Improvements in tracking and patrol capacity, introduction of AIS and VMS more widely, and various other MCS efforts, can help to reduce incidence of conflicts. However, until there is the means in fisheries law and regulations to control new construction of artisanal fishing boats this issue will likely remain.

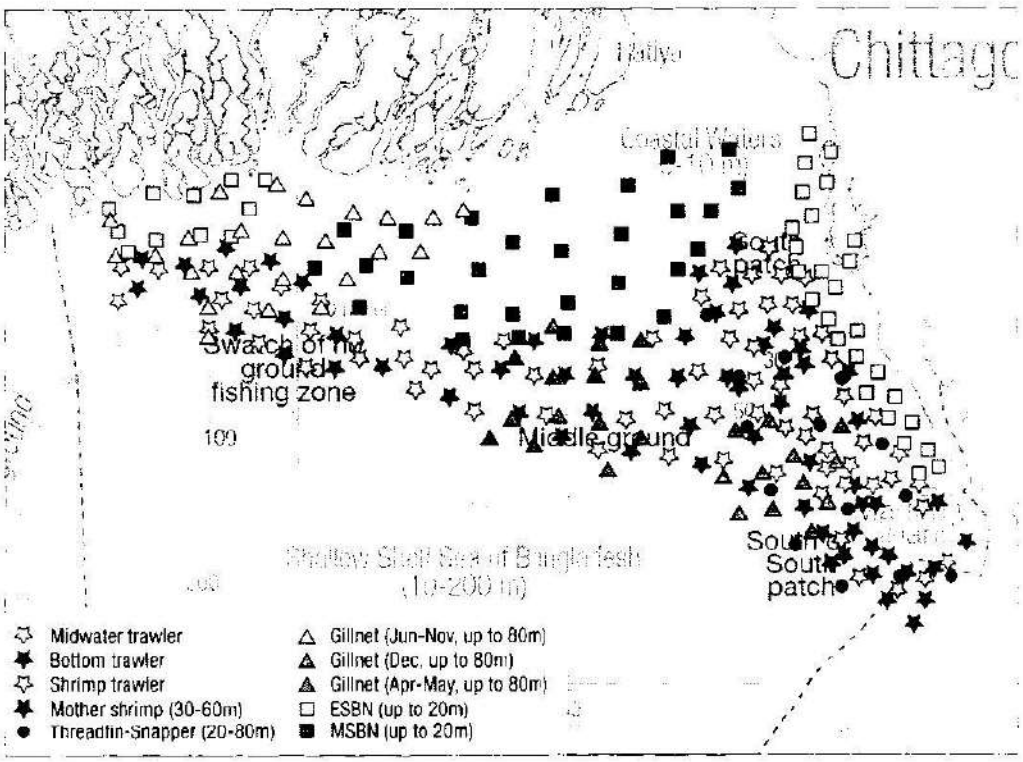

Figure 6 Fishing grounds of various gear sectors operating from Chattogram and Cox's Bazar showing overlap areas of potential conflict.

Ecosystem considerations

There is major concern over the non-selective over-exploitation of fishery resources by the multitude of fishing gears being used, especially the bottom trawlers in the industrial sector and the estuarine set bag nets (ESBN) in the artisanal sector. There is limited or no regulation to control bycatch and the mesh size regulations are widely ignored, resulting in large catches of juveniles in many cases.

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The threat to resources near St. Martin's Islands stemming from many vessels, both industrial and artisanal, fishing within easily damaged coral reef and sea grass ecosystems is another concern.

Ecosystem considerations raised in the survey reports from RV Meen Shandhani (Fanning et al 2019) noted that there has been a replacement of larger, slower-growing and slower-reproducing groups such as Indian salmon with small-sized, fast-growing and fast-reproducing species such as sardines and scads. In a healthy ecosystem, these species are the forage base supporting the larger predatory species. Thus the increase of small species reflects a significant alteration to the ecosystem structure which may or may not be reversible in a reasonable time period. The ability to rebuild the stocks of larger and more valuable species can be significantly impaired if such an ecosystem shift has become well established. The bycatch of demersal species juveniles in the growing industrial fishery on small pelagic species will further impair the ability to rebuild the high-values stocks.

High-priority issues

There exists many challenges confronting the marine fisheries but given the prevailing state of marine fish catch, including species composition and serious scarcity of economically high valued species, the high priority issues are:

- Over-capacity in the industrial fleet leads to over-exploitation of the resources and reduces the economic performance of the sector
- Prevalence of Illegal, Unreported and Unregulated (IUU) fishing by both national and foreign vessels
- Ineffective MCS system in place-dearth of human resources, logistics and enforcement capacity
- Inadequate good governance including non-compliance with existing rules and regulations, exacerbated by the top-down decision-making process and lack of co-management by resource users
- Lack of economic returns due to post-harvest losses and limited value chains and marketing opportunities
- Lack of food safety and HACCP due to inadequate preservation and postharvest measures.

These high priority issues have been identified from stock status reports, various other reports of DoF, and were expressed in the opinions and recommendations from consultations with industrial fishing vessel owners, associations, skippers & crews, and concerned stakeholders of both public and private sector. They reflect the prevailing gaps in enforcement capacity, inadequate rules and regulations, lack of systematic and historical data. Addressing these issues means addressing the underlying causes (Table 9), including the lack of both human logistical capacity.

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Table 9 Issues affecting the industrial trawling sector of Bangladesh

High priority issues	Cause of Issues
Overfishing and overcapacity of industrial FVs seriously impact on marine fisheries stocks.	• Large number of industrial vessel capturing marine resources without knowing stock size; • Weak control to maintain size or number of fishing gears being operated; • High intensity of both growth and recruitment overfishing with non-compliance of marine fisheries rules and regulations; • Ineffective enforcement of existing marine fisheries Act and regulations; • Degradation of habitat and ecology due to indiscriminate fishing and pollution;
Ineffective MCS procedure impedes in controlling IUU fishing.	• Inadequate coordination among agencies involved in MCS procedure; • Absence of effective monitoring and surveillance schemes in place (monitoring & inspection at sea and port, no observer scheme); • Lack of communications systems to share monitoring information among maritime domains for joint operation; • Inadequate authority and deterrence in dealing with entry of foreign fishing vessels with records of IUU fishing;
Other threats and issues	
Destructive fishing practices	• Use of small mesh sizes in fishing gears both at cod-end and beyond- most trawler also use bag net over cod-end; • Destroy large quantities of juveniles of high valued larger commercial species • Destruction of jatka (juvenile hilsa) at high intensity in recent yrs. by mid-water trawlers mostly operated in low depth <40 m; • Demand for low value/trash fish for drying/fish meal
Conflicts between artisanal and commercial fishers'	• Overcapacity & intense fishing effort leads to competition for depleted resources; • Infringement of fishing zones by industrial trawlers & non-compliance of legal regulations; • No control to build artisanal boats and engaging to fish;
Inadequate or absence of good governance	• Complete absence of co-management arrangement in the management of common pool resources; • No consultation and agreed decision among various marine fisheries associations for sustainability of resources; • Insufficient institutional research and development activities on marine fisheries; • Lack of coordination among different marine domain for proper enforcement; • DoF does not have any enforcement unit to oversight MCS;
Loss of food quality and value of landed catch due to lack of due on-board post-harvest handling and preservation technology.	• Quality deteriorates of landed catch on-board exposed to sun; • Majority of industrial trawlers do not have provision to use refrigerated sea/chilled water to maintain cool chain during sorting and bagging or kept in trays or basket; • More than 80% catch by all Industrial FVs kept in polyethylene bag as lump before transfer to freezer for preservation that takes more than 24 hrs. to freeze whole product; • Proper hygiene and sanitation and HACCP implementation is not maintained by most of the FVs; as a result quality deteriorates and price become less during auction. • Crews of FVs are not well trained to maintain personnel hygiene facing occupational hazards.
Asymmetry and Inadequate fisheries data & information	• Inadequate human capacity of technical and scientific • staff involved in data collection and analyses; • Inadequate time and effort allocated to collect and analyse data and information required for Key KPIs to monitor FMP;
Inadequate fisheries management capacity	• Lack of an effective fisheries management & enforcement unit in DOF; • Insufficient logistics e.g. patrol boats, vehicles etc. to strengthen management measures;

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